WORKED SOLUTION

STEP I 2004, Question 1

Pure • Surds and Cube Roots

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q1

The key idea

Numbers of the form $a + b\sqrt{n}$ with integer a, b behave like a little number system: multiplying two of them gives another of the same form, and two such numbers are equal exactly when their rational parts agree and their $\sqrt{n}$ parts agree. Part (i) is a warm-up in cubing such a number; part (ii) runs the process backwards by comparing parts. Part (iii) is a quadratic in x^3, and its roots turn out to be exactly the cubes found in (ii).

Solution

Part (i)

By the binomial theorem, with $(2\sqrt{5})^2 = 20$ and $(2\sqrt{5})^3 = 40\sqrt{5}$,

$$\begin{aligned}\left(3+2\sqrt{5}\right)^3 &= 3^3 + 3\cdot 3^2\cdot 2\sqrt{5} + 3\cdot 3\cdot 20 + 40\sqrt{5}\\ &= 27 + 54\sqrt{5} + 180 + 40\sqrt{5}\\ &= 207 + 94\sqrt{5}.\end{aligned}$$

So $a = 207$ and $b = 94$.

Part (ii)

The same expansion with $\sqrt{2}$ in place of $\sqrt{5}$ gives, for any c and d,

$$\begin{aligned}\left(c-d\sqrt{2}\right)^3 &= c^3 - 3c^2d\sqrt{2} + 3c\cdot 2d^2 - 2d^3\sqrt{2}\\ &= \left(c^3 + 6cd^2\right) - \left(3c^2d + 2d^3\right)\sqrt{2}.\end{aligned}$$

Since the real cube root is unique, we need $(c - d\sqrt{2})^3 = 99 - 70\sqrt{2}$.

Ask yourself Why may we compare the rational parts and the $\sqrt{2}$ parts separately?

If $p+q\sqrt{2}=r+s\sqrt{2}$ with p,q,r,s integers and $q\neq s$, then $\sqrt{2}=(p-r)/(s-q)$ would be rational, which it isn't. So $q=s$, and then $p=r$. Hence we need

$$c^3+6cd^2=99, \qquad 3c^2d+2d^3=70.$$

Since c and d are positive, the first equation forces $c^3<99$, so $c\leq 4$. Trying each value in $6cd^2=99-c^3$:

- $c=1$: $6d^2=98$, no integer solution;
- $c=2$: $12d^2=91$, no;
- $c=3$: $18d^2=72$, so $d=2$;
- $c=4$: $24d^2=35$, no.

The only candidate is $c=3$, $d=2$, and it satisfies the second equation too: $3\cdot 9\cdot 2+2\cdot 8=54+16=70$. Therefore

$$\sqrt[3]{99-70\sqrt{2}}=3-2\sqrt{2}, \qquad c=3,\ d=2.$$

(As a sanity check on signs, $2\sqrt{2}\approx 2.83<3$, so $3-2\sqrt{2}$ is positive, matching the fact that $99-70\sqrt{2}\approx 0.005$ is positive.)

Part (iii)

Put $y=x^3$. The equation becomes $y^2-198y+1=0$, so

$$y=\frac{198\pm\sqrt{198^2-4}}{2}=99\pm\sqrt{99^2-1}=99\pm\sqrt{9800}=99\pm 70\sqrt{2},$$

using $9800=4900\cdot 2$. Both values are positive and real.

From part (ii), $99-70\sqrt{2}=(3-2\sqrt{2})^3$. For the other root, notice that the product of the two roots of the quadratic is 1, so

$$99+70\sqrt{2}=\frac{1}{99-70\sqrt{2}}=\frac{1}{(3-2\sqrt{2})^3}=\left(3+2\sqrt{2}\right)^3,$$

since $(3-2\sqrt{2})(3+2\sqrt{2})=9-8=1$. (Equally, changing the sign of d in the expansion of part (ii) gives $(3+2\sqrt{2})^3=99+70\sqrt{2}$ directly.)

Each real number has exactly one real cube root, so each value of y gives exactly one real x. The two real solutions are

$$x=3-2\sqrt{2} \qquad \text{and} \qquad x=3+2\sqrt{2}.$$

The other four solutions of the sextic are complex: the remaining cube roots of each y.

Checks

- Numerically, $(3+2\sqrt{5})^3 \approx 417.19$ and $207+94\sqrt{5} \approx 417.19$.
- $3+2\sqrt{2} \approx 5.8284$, and $x^6 - 198x^3 + 1$ at this value is zero to rounding error; the same holds at $3-2\sqrt{2} \approx 0.1716$.
- The two answers to (iii) multiply to 1, as they should: if x is a root then so is $1/x$, because dividing the equation by x^6 gives the same equation in $1/x$.

Common slips

- Forgetting that $(d\sqrt{2})^3 = 2\sqrt{2}\,d^3$, not $\sqrt{2}\,d^3$, when expanding.
- Finding $c=3$, $d=2$ from the first equation but not checking the second.
- Giving six "solutions" in (iii): only one cube root of each positive number is real.

Beyond STEP

The set of numbers $a+b\sqrt{2}$ with a, b integers is a ring, written $\mathbb{Z}[\sqrt{2}]$, and $3+2\sqrt{2}$ is a *unit* in it: it has an inverse, $3-2\sqrt{2}$, in the same ring. Units like this are tied to Pell's equation $a^2 - 2b^2 = \pm 1$ (note $99^2 - 2\cdot 70^2 = 1$), and every unit of $\mathbb{Z}[\sqrt{2}]$ is $\pm(1+\sqrt{2})^n$. For more, see Davenport, *The Higher Arithmetic*, or Allenby, *Rings, Fields and Groups*.

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